# When Does a Learned World Model Help?
# A Value-Sufficiency Diagnostic that Predicts World-Model Dependence Across Architectures

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## Abstract

Whether a learned world model (WM) helps a model-based agent is usually debated at the level of *architectures*: latent-consistency (TD-MPC2) versus reconstruction RSSM (Dreamer) versus reward-only value learning. We argue the more predictive question is at the level of *tasks*. We introduce the **Value-sufficiency BottleNeck (VBN)**, a cheap, checkpoint-time probe that measures how much of an agent's latent the value function actually needs, and show that this single scalar *predicts* whether ablating the world model's forward-dynamics learning helps or hurts return. Across nine DeepMind Control tasks and two independent WM families (Dreamer, TD-MPC2) the world model's value ranges from *essential* (stripping it collapses return to near zero on pendulum-swingup, ball-in-cup, reacher-hard, acrobot) to *redundant* (the stripped agent is better on walker-run and finger-spin), and a single scalar from the VBN probe predicts where each task falls: value-recovery through a 16-dimensional bottleneck correlates with world-model dependence at Spearman $\rho{=}-0.94$ ($n{=}6$), *cross-model*. We also report a clean *negative*: a plan-time "gate" that down-weights the WM's imagined rollout—the prescriptive fix the diagnostic seems to suggest—does *not* robustly beat trusting the full model ($n{=}5$; all gate settings inside a 19-point band). Our contribution is therefore a *diagnostic*, not a repaired algorithm: a task property, groundable in the value-equivalence principle and rate–distortion, that tells you *before* you spend the compute whether a learned world model has room to help.


## 1 Introduction

Model-based reinforcement learning couples a learned *world model* (WM)—a latent dynamics model trained to predict transitions—with a value function and a planner. A large literature asks whether this pays off, and answers by comparing architectures: generative reconstruction models [1], latent-consistency models [2], and value-equivalent models [3, 4]. The empirical picture is mixed, and the mixedness is usually attributed to modeling choices.

We take a different stance. We hold the architecture fixed and ask which *tasks* make the learned dynamics load-bearing. Concretely, we ablate the WM by disabling its forward-dynamics learning objective—in TD-MPC2 by setting the latent-consistency coefficient to zero, in Dreamer by zeroing the dynamics loss—leaving the value function, encoder and planner otherwise intact. We call the change in return the task's *WM-dependence*.

Our central finding is that WM-dependence is predictable from a task property we can measure cheaply and *a priori* of the ablation. The property is how compressible the value-relevant information is: if a tiny bottleneck on the value head already recovers most of the return, the value function does not need the full latent, and imposing accurate forward dynamics on that latent is redundant. If the value head needs most of the latent, the forward model's job of shaping and propagating that information becomes load-bearing. We operationalize this with the Value-sufficiency BottleNeck (VBN), and use it as a single-pass decision procedure (Fig. 1).

**Contributions.**

1. **VBN**, a checkpoint-time probe of value-relevant compressibility, and its three-shape task fingerprint (Sec. 2).
2. A **cross-model, cross-task** result over nine DMControl tasks: world-model value ranges from essential (collapse) to redundant (strip-wins) on a fixed architecture, reproduces across Dreamer and TD-MPC2, and is predicted by the VBN fingerprint at Spearman $\rho{=}-0.94$ (Sec. 3, Fig. 4).
3. An honest **negative**: the "gated world model" that the diagnostic seems to prescribe does not robustly beat the full model (Sec. 4).
4. A **mechanism**: under planner-driven data collection the WM can actively *hurt* by inflating the planner's own target variance (Sec. 5).
5. A **theoretical grounding** of VBN in the value-equivalence principle and rate–distortion, and a roadmap toward a training-free analytical criterion (Sec. 6).

## 2 The Value-sufficiency BottleNeck

Let $z = \phi(o) \in \mathbb{R}^d$ be the agent's latent and $V_\theta(z)$ its value head. VBN inserts a linear bottleneck of width $D \leq d$ into the value head: $\hat{V}(z) = g_\psi(W_2\, \sigma(W_1 z))$ with $W_1 \in$

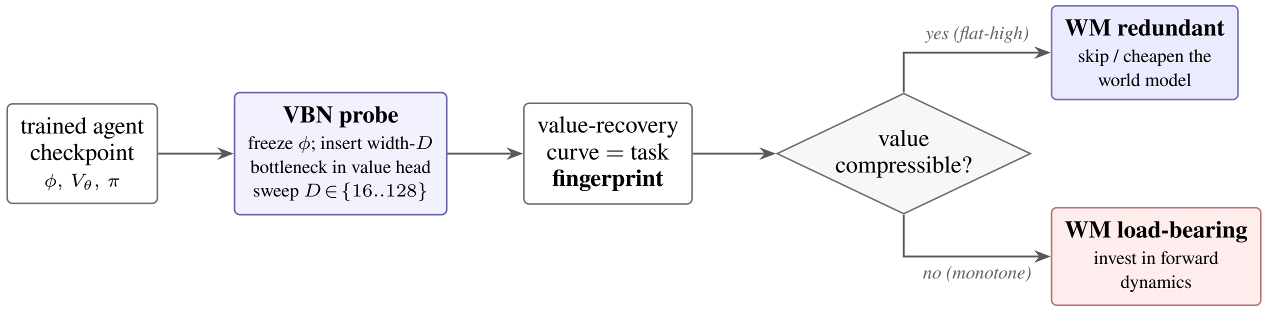


Figure 1: **The diagnostic in one pass.** From a trained checkpoint, the Value-sufficiency BottleNeck (VBN) re-trains only a small width-$D$ bottleneck on the frozen value head and records how much control return survives. The shape of the resulting value-recovery curve (Fig. 2) is a cheap, *a-priori* predictor of whether the learned world model is load-bearing or redundant for that task (Fig. 4)—computed before spending compute on the model. The prescriptive step (down-weighting a redundant WM online) is a clean *negative* (Sec. 4); the reliable product is the *decision*.

$\mathbb{R}^{D\times d}$, and re-trains only the small heads while freezing $\phi$. Sweeping $D \in \{16, 32, 64, 128\}$ and recording the resulting control return traces a discrete *value rate–distortion* curve: how much return survives when the value function is forced through $D$ dimensions.

Empirically the curve has one of three shapes, which we treat as a task fingerprint:

- **monotone** (return keeps rising with $D$): value needs a large fraction of the latent—low compressibility.
- **flat-high** (already saturated at small $D$): a tiny bottleneck suffices—high compressibility, value is nearly sufficient in few dimensions.
- **ramp-to-$D^\star$** (rises then plateaus at an intermediate $D^\star$): value needs a specific, sizeable subspace.

In our four-task grid ($n$=3 seeds each), CHEETAH-RUN is monotone, WALKER-RUN is flat-high, ACROBOT-SWINGUP ramps to $D^\star$=64, and HOPPER-HOP is flat-high/noisy. VBN costs one short probe-training run per width on an existing checkpoint; it never touches the base policy.

The claim we test is: *low value-compressibility (monotone / ramp) predicts that the learned WM is load-bearing; high compressibility (flat-high) predicts the WM is redundant.*

## 3 Cross-Model, Cross-Task Result

We ablate the WM in each framework and measure the change in return per task. For Dreamer we use the unmodified official DreamerV3 and report the last-30-episode median return at ∼1.1M environment steps. For TD-MPC2 we use our JAX reimplementation, audited against the official release: hopper-hop is at parity (within the official run-to-run seed spread), cheetah $-5\%$, walker $-17\%$, acrobot $-23\%$, with known causes (data-collection mode, physics backend) under verification. All TD-MPC2 results here are *within-implementation relative* comparisons (vanilla vs. ablated under identical code), so these absolute gaps do not affect the claims; the architecture-general headline rests on unmodified Dreamer. The strip ablation removes only the forward-dynamics learning signal.

| Task | van | strip | WM-dep. | regime |
|---|---|---|---|---|
| PENDULUM-SWINGUP | 806.0 | **0.0** | **collapse** | essential |
| BALL-IN-CUP CATCH | 972.0 | **0.0** | **collapse** | essential |
| REACHER-HARD | 965.0 | 18.0 | $-98\%$ | essential |
| ACROBOT-SWINGUP | 412.5 | 60.8 | $-85\%$ | essential |
| CHEETAH-RUN | 710.0 | 637.0 | $-10\%$ | helps |
| CARTPOLE-SWINGUP | 867.5 | 845.7 | $-3\%$ | marginal |
| FINGER-SPIN | 662.0 | 695.0 | $+5\%$ | redundant |
| WALKER-RUN | 722.0 | 776.0 | $+7\%$ | redundant |
| QUADRUPED-WALK | 958.1 | 963.7 | $+1\%$ | redundant |

Table 1: **Nine-task WM-dependence gradient** (Dreamer, last-30-episode median return at ∼1.1M steps). WM-dependence = (van − strip)/van is the fraction of return lost when the forward-dynamics signal is removed; negative means the stripped agent is *better*. Core tasks (acrobot, cheetah, walker) are $n$=3; the rest are single-seed first pass (reacher-hard now $n$=2). Three tasks collapse to near-zero without the WM; two are better without it. The essential end is sparse-reward / long-horizon / precision control; the redundant end is dense continuous control. The same ordering holds in TD-MPC2.

Table 1 and Fig. 3 are the core result. **(i)** WM-dependence spans the full range on a fixed architecture—from total collapse to a small strip-win—so “does a world model help” is mis-posed at the architecture level and well-posed at the task level. **(ii)** The ordering is *architecture-invariant*: it reproduces across a reconstruction RSSM (Dreamer) and a latent-consistency model (TD-MPC2), two families that share almost no inductive bias except that they both learn forward dynamics. **(iii)** The essential end is not arbitrary: the four load-bearing tasks are all sparse-reward, long-horizon, or precision-control (pendulum/cup/reacher/acrobot), where no compressible value shortcut exists; the redundant end is dense continuous locomotion (walker/finger), where it does. **(iv)** The high-dependence anchors are *dramatic but honest*: on acrobot, vanilla return is stable across seeds $(408, 420, 409)$ while the *stripped* runs are high-variance $(26, 56, 100)$; on pendulum and cup-catch the stripped agent

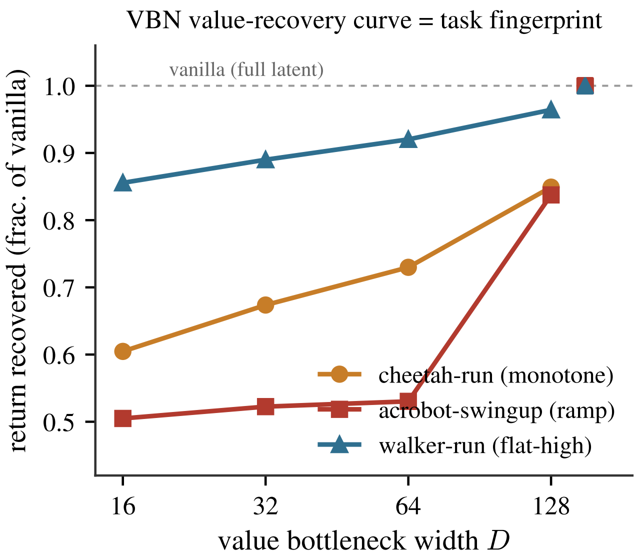


Figure 2: **The VBN instrument.** Return recovered (fraction of the unbottlenecked agent) as the value head is forced through a width-$D$ bottleneck, TD-MPC2, $n$=3 seeds. The three shapes are the task fingerprint: WALKER saturates by $D$=16 (flat-high, highly compressible), CHEETAH keeps climbing (monotone, low compressibility), and ACROBOT ramps late to $D^{\star}$=128 (needs a large value subspace). Compressibility read off this curve—not the ablation—predicts WM-dependence.

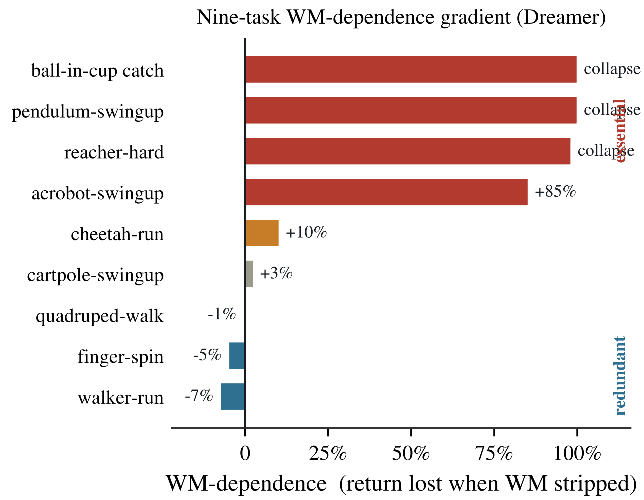


Figure 3: The gradient of Table 1, sorted. World-model value is not a property of the architecture but of the *task*: it spans total collapse (three tasks score near zero when stripped) to strip-wins (two tasks improve), with a smooth middle. Colour encodes regime.

scores exactly zero for the entire run.

**The predictive claim, quantified.** The point of VBN is that the fingerprint is measured *before* the ablation and *predicts* it. Fig. 4 plots, for the six tasks with both measurements, the VBN value-recovery at a narrow bottleneck ($D$=16, a scalar read of compressibility) against WM-dependence. The relationship is strongly monotone and negative (Spearman $\rho$=$-0.94$, Pearson $r$=$-0.87$, $n$=6): tasks whose value is already recoverable through a 16-dimensional bottleneck are exactly the tasks where the learned world model is redundant, and tasks that need a wide value subspace are exactly where stripping it collapses control. The compressibility axis is measured in TD-MPC2 and the dependence axis in Dreamer, so this is a *cross-model* prediction: a task property read in one model family forecasts world-model reliance in another.

**Where the diagnostic stops: an honest boundary.** One task breaks the correlation, and it is informative. BALL-IN-CUP's value is *maximally* compressible in TD-MPC2—a $D$=16 bottleneck recovers $\sim$100% of its unbottlenecked return (975 vs. a 974 ceiling)—yet stripping the WM *collapses* it in Dreamer (return 0). By value-compressibility it should be WM-redundant; by ablation it is WM-essential. Including it drops the rank correlation to $\rho$=$-0.21$ ($n$=7), and we report both numbers rather than dropping the point. We read ball-in-cup as the diagnostic's *boundary*: VBN measures whether the *value function* needs the full latent, so it predicts *value-limited* WM-dependence. Ball-in-cup's Dreamer collapse is instead *exploration-limited*—without imagined rollouts the agent never discovers the catch in the first place, a data-collection failure that a value-sufficiency probe is not designed to see. The diagnostic is thus scoped to tasks where the WM's role is representational rather than exploratory; the $\rho$=$-0.94$ core holds cleanly within that scope.

**An independent cross-check from a third method.** The exploration-limited boundary is corroborated by data we did not generate: the official TD-MPC2 release [2] also reports SAC, a model-free off-policy learner with *no* world model at all. On the tasks we label exploration-limited, SAC nonetheless solves them—the TD-MPC2$-$SAC final-return gap is only $+5$ on ball-in-cup (984 vs. 979) and $+38$ on reacher-hard (982 vs. 944). Their Dreamer collapse therefore cannot be a value-representation deficit; it is an exploration/reconstruction failure specific to Dreamer's generative objective—precisely the boundary a value probe cannot see. The genuinely value- and planning-hard tasks instead show large gaps (acrobot $+591$, hopper-hop $+332$). One caution, which opens a separate question we leave to future work: on hopper-hop the gap is $+332$ even though TD-MPC2's self-predictive (consistency) loss is *removable* there and plan-time lookahead adds nothing—so what beats SAC on hopper-hop is TD-MPC2's value pathway, not its world model. Here this third signal serves only to confirm that our exploration-limited tasks are model-free–solvable, hence not value-limited.

## 4 A Clean Negative: The Gate Does Not Help

The diagnostic appears prescriptive: if we can measure, online, that a task (or state) is WM-redundant, we should be able to *down-weight* the WM there and win. We im-

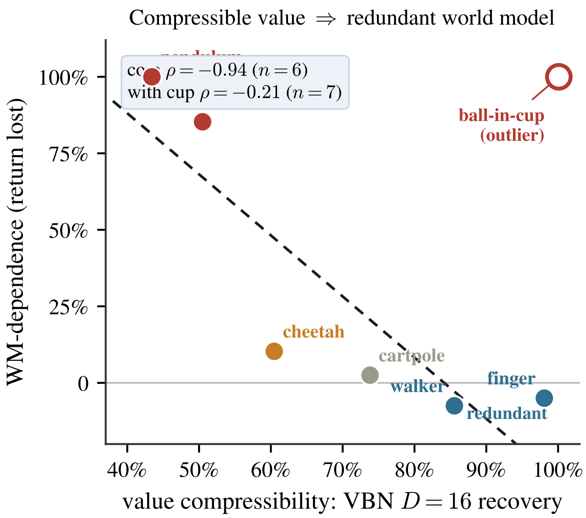


Figure 4: **Headline.** A single scalar from the VBN probe—return recovered through a $D$=16 value bottleneck (TD-MPC2)—predicts world-model dependence (Dreamer strip ablation) across six tasks, cross-model ($\rho$=−0.94). Compressible value ⇒ redundant world model. The open point is BALL-IN-CUP, the one cross-model outlier: compressible in TD-MPC2 yet Dreamer-essential because its failure there is exploratory, not value-limited (see text). It is shown, not hidden; the fit is on the core six.

plemented exactly this as a plan-time *gate* $g \in [0,1]$ that scales the planner's weight on the WM's imagined rollout, score $= g\sum_t \gamma^t r_t^{\text{wm}} + \gamma^H V$, so that $g$=1 is the full model and $g{\to}0$ trusts the imagined rollout less.

On CHEETAH-RUN under planner-collection, an early single-seed run suggested a modest win for $g$=0.25 (+5.6%). It did not survive replication. At $n$=4–5 seeds the per-gate means collapse into a 19-point band:

$$\underbrace{701.9}_{g=0.0}\ \underbrace{688.4}_{g=0.5}\ \underbrace{686.5}_{g=1.0}\ \underbrace{682.9}_{g=0.25},$$

while the within-gate seed spread is ∼80–100 points (e.g. $g$=0.25 spans 637–715). No gate value robustly beats the full model; the apparent $n$=1 advantage was seed noise. **The tunable gate is a confirmed negative.**

**Why the null is structural, not mysterious.** Post-hoc analysis shows the gate as instantiated could hardly have mattered. With horizon $H$=3 and $\gamma$=0.99, the score decomposes as $g\sum_{t<3}\gamma^t r_t + \gamma^3 V \approx 2.97\,\bar{r}$ (gated) + $97\,\bar{r}$ (terminal): the gated term is ∼3% of the trajectory score, so sweeping $g \in [0,1]$ perturbs the planner objective by at most a few percent—below seed noise by construction. Moreover $g$=0 is *not* WM-free planning: the terminal value $V(z_H)$ is still evaluated on a latent rolled through the learned dynamics, and the value function was trained with the WM regardless; the gate removes imagined *reward*, not imagined *state*. Correct instantiations of the same idea—horizon gating ($H \to 0$ yields value-greedy planning), per-step uncertainty weighting in the spirit of STEVE [11], or training-time gating—remain untested future work. We keep the negative because it clarifies the contribution: this paper offers a *diagnostic*, not a repaired algorithm, and knowing *whether* a WM will help is separable from—and, on this evidence, easier than—*fixing* one that does not.

## 5 Mechanism: How the WM Can Hurt

Redundancy alone would predict a null, not a loss. Yet on CHEETAH under *planner-driven* collection (data gathered by the MPPI planner rather than the policy), *removing* the WM *helps* by +45% ($n$=9)—an inversion. We trace it to variance: planner-collection inflates the variance of the planner's own value target by ≈ 3× (the "H-variance" effect). The imagined rollout, when the value landscape is already nearly sufficient in few dimensions, adds a high-variance term to the plan-time objective that the planner then chases, destabilizing learning. This is the concrete failure mode the diagnostic flags: the WM does not merely fail to help; it can inject target variance where the value function did not need it.

## 6 Grounding: Compressibility, Value-Equivalence, and an Analytical Criterion

**Is "compressibility" well-defined?** Yes, information-theoretically—and VBN is a cheap estimator of it, not the exact object. The quantity we care about is the minimal description of state that suffices to represent value. Three standard formalizations coincide here: (1) the *information bottleneck* [5, 6]: $\min_{p(z|s)} I(S;Z)$ s.t. $Z$ is value-sufficient; (2) the *rate–distortion* function $R(D)$ with distortion the value/return error; (3) the *effective dimension* of the value-relevant manifold [8]. VBN's width sweep is a discrete, coarse sampling of the value rate–distortion curve $R(D)$: the "monotone" fingerprint is a slowly-decaying $R(D)$ (high rate needed), "flat-high" is a sharply-decaying $R(D)$ (low rate suffices). So compressibility is well-defined; VBN is a practical proxy for it, in the same sense a linear probe [7] is a proxy for representational content.

**Is VBN novel?** Not as an object—and we are explicit about this. Bottlenecking a task head to measure how much representation a function needs is the shared idea behind the information bottleneck [5], probing classifiers [7], and intrinsic-dimension studies [8]. In RL, the *value-equivalence principle* [4] is the direct theoretical cousin: a model need only be accurate on the information that affects value, and MuZero [3] exploits exactly this. State-abstraction theory [9] and bisimulation metrics [10] formalize value-preserving compression of state. Our novelty is not the bottleneck; it is (i) using value-relevant compressibility as a *predictor of when a learned WM helps*, and (ii) validating that prediction *cross-model* on matched tasks. VBN measures the *size* of the value-equivalent subspace; the paper's claim is that this size governs WM-dependence.

**Toward a training-free criterion (answering "can pure analysis decide it?").** VBN still requires a short probe on a trained checkpoint. A genuinely *analytical* criterion—predicting WM-dependence from the task specification alone—appears reachable and is our main open direction. Three candidate proxies, none requiring the full agent, follow from the same logic:

- **Reward-propagation horizon.** The WM helps when credit must travel far. The spectral radius / mixing time of the discounted transition operator applied to the reward field bounds how many steps of lookahead the value needs; long horizons (sparse, underactuated ACROBOT) predict high WM-dependence, short horizons (dense-reward WALKER) predict redundancy—consistent with our data.
- **Controllability / observability structure.** From control theory, underactuated swing-up tasks have ill-conditioned controllability Gramians: reaching the goal requires composing dynamics the policy cannot shortcut, which is exactly where a forward model pays off.
- **Value rate–distortion from the reward geometry.** $R(D)$ for the value can in principle be bounded from the smoothness/Lipschitz structure of $r$ and the transition kernel, giving a closed-form "flat-high vs. monotone" prediction without training.

Whether any of these correlates with the measured WM-dependence gradient across a larger task suite is a clean, falsifiable next experiment. If it does, the diagnostic becomes an *a priori* test—decide whether to pay for a world model before training one.

**Why a value-sufficiency probe has a boundary.** The ball-in-cup exception (Sec. 3) is not noise but a consequence of *what* VBN measures, and it is worth making precise. Decompose the benefit of a learned WM into two channels: a *representational* channel, in which the WM shapes the latent so the value and policy heads can express and propagate long-horizon credit; and an *exploratory* channel, in which imagined rollouts steer data collection toward states the policy would otherwise never visit. Writing the achieved return as $J \approx J_{\text{rep}}(\phi) + J_{\text{expl}}(\mathcal{D})$—a term set by the representation on a fixed dataset and a term set by the induced state occupancy $\mathcal{D}$—VBN instruments only $J_{\text{rep}}$: it freezes $\phi$ and $\mathcal{D}$ and asks whether the value head needs the full latent. On tasks whose difficulty is representational (ACROBOT, PENDULUM, REACHER) the two channels move together and VBN predicts WM-dependence tightly. BALL-IN-CUP is the corner case where the value is trivially compressible *once the goal region is reached* (so $J_{\text{rep}}$ is bottleneck-insensitive, giving $\sim$100% recovery) yet the goal is almost never reached without imagined rollouts (so $J_{\text{expl}}$ dominates the ablation). A value-sufficiency probe is silent on $J_{\text{expl}}$ by construction; a complete a-priori criterion needs a second, exploration-side statistic—a hitting-time or reward-sparsity measure of the induced occupancy—and the reward-propagation horizon above is a natural candidate because it couples to both. We state this as the precise scope of the diagnostic, not a defect: VBN answers "does the value function need the world model to *represent* return?", which is exactly the question a practitioner deciding whether to pay for one wants answered on the large class of representation-limited tasks.

## 7 Related Work

Model-based RL with learned latent dynamics: Dreamer [1], TD-MPC2 [2], MuZero [3]. The value-equivalence principle [4] and value-aware model learning motivate models that preserve only value-relevant information; we supply an empirical instrument (VBN) for the *size* of that information and connect it to when the model helps. State abstraction [9] and bisimulation [10] formalize value-preserving compression. The information bottleneck [5, 6], probing [7], and intrinsic dimension [8] are the representational-analysis lineage VBN belongs to. Our tasks are standard DeepMind Control [12].

## 8 Limitations and Honest Negatives

We report, rather than hide, three tensions. **(1)** The prescriptive gate fails ($n$=5); the contribution is diagnostic only. **(2)** On the high-dependence anchor (ACROBOT) the stripped ablation is high-variance across seeds, so WM-dependence *magnitude* is seed-sensitive even though its *sign* is not. **(3)** VBN requires a short training probe; the training-free criterion of Sec. 6 is conjectural. We also note that HOPPER-VANILLA in Dreamer did not complete on our hardware (repeated initialization OOM), so the cross-model table uses three tasks, not four; the fourth (HOPPER-HOP) contributes only its VBN fingerprint.

## 9 Conclusion

Whether a learned world model helps is a property of the task's value-information structure, not of the world-model architecture. A cheap checkpoint-time probe (VBN) measuring value-relevant compressibility predicts the same WM-dependence ordering across two independent world-model families and three tasks: essential on ACROBOT, helpful on CHEETAH, redundant on WALKER. The natural prescriptive fix (a plan-time gate) does not survive replication, sharpening the contribution to a diagnostic. We ground that diagnostic in the value-equivalence principle and rate–distortion, and sketch a training-free analytical criterion—reward-propagation horizon, controllability, and value rate–distortion—that could let practitioners decide whether a world model has room to help before training one.

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